ENGLISH

Datasheet
5MHz/3MHz Function Generator

Stock No. :	Model :
123-3554	GFG-8216A (3MHz)
123-3555	GFG-8219A (3MHz)
123-3556	GFG-8255A (5MHz)



GFG-8216A (3MHz)



GFG-8219A (3MHz)



GFG-8255A (5MHz)

FEATURES

- Frequency Range : 0.3Hz ~ 3MHz (GFG-8216A/8219A)
- Frequency Range : 0.5Hz ~ 5MHz (GFG-8255A)
- Waveforms : Sine, Triangle, Square,Ramp, TTL and CMOS Output
- External Voltage Controlled Frequency (VCF) Function
- Duty Cycle Control with Signal Inversion Capability
- Variable DC Offset Control
- Two-Steps (-20dBx2) and Variable Attenuator
- Built-in 6 Digits Counter with INT/EXT Function up to 150MHz
- LIN/LOG Sweep Mode (GFG-8219A/8255A)
- INT/EXT AM/FM Modulation (GFG-8219A/8255A)
- GCV Output for Synchronization (GFG-8219A/8255A)

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The GFG-8200A Series Function Generators are embedded with standard functions such as TTL, CMOS and RAMP outputs, VCF (external voltage control frequency) and a 6-digit frequency counter. All these integrated functions and the user-friendly operation are made to accommodate applications in audio response testing, vibration testing, servo system evaluations, and ultrasound applications.

In addition, the GFG-8255A and 8219A models include logarithmic and linear sweep, GCV (Generator Controlled Frequency) output and AM/FM modulation. The GFG-8200A Series' models span two frequency bandwidths; the GFG-8216A/8219A at 0.3Hz ~ 3MHz and the GFG-8255A at 0.5Hz ~ 5MHz.

SPECIFICATIONS

MODEL	GFG-8216A/8219A	GFG-8255A
MAIN		
Frequency Range	0.3Hz ~ 3MHz (7 Range)	0.5Hz ~5MHz (7 Range)
Amplitude	> 10Vpp (into 50Ω load)	> 10Vpp (into 50Ω load)
Impedance	50Ω ±10%	50Ω ±10%
Attenuator	-20dB ± 1dB x 2	-20dB ± 1dB x 2
DC Offset	<-5V ~ >5V (into 50Ω load)	<-5V ~ >5V (into 50Ω load)
Duty Control	20% ~ 80%,maximum to 1MHz (continuously adjustable)	20% ~ 80%,maximum to 1MHz (continuously adjustable)
Display	6 digits LED display *GFG-8215A does not have a display	6 digits LED display
Range Accuracy	±5%+1Hz (at 3 positions) *only for GFG-8215A	
SINE WAVE		
Distortion THD	≤ 1%, 0.3Hz ~ 200kHz ≤ -35dB below fundamental in all range (MAX. to 1/10 level)	≤1%, 0.5Hz ~ 100kHz ≤-30dB below fundamental in all range (MAX. to 1/10 level)
Flatness	< 0.3dB, 0.3Hz ~ 300kHz < 0.5dB, 300kHz ~ 3MHz	≤0.3dB, below 500kHz ≤1dB, below 5MHz
TRIANGLE WAVE		
Linear	≥ 98%, 0.3Hz ~ 100kHz ≥ 95%, 100kHz ~ 3MHz	≥ 98%, 0.5Hz ~ 100kHz ≥ 95%, 100kHz ~ 5MHz
SQUARE WAVE		
Symmetry	± 2%, 0.3Hz ~ 100kHz	± 2%, 1Hz ~ 100kHz
Rise or Fall Time	≤ 100ns at maximum output (into 50Ω load)	≤ 50ns at maximum output (into 50Ω load)
CMOS OUTPUT		
Level	4Vpp±1Vpp ~ 14.5Vpp±0.5Vpp adjustable	4Vpp±1Vpp ~ 14.5Vpp±0.5Vpp adjustable
Rise or Fall Time	≤ 120ns	≤ 120ns
TTL OUTPUT		
Level	≥ 3Vpp	≥ 3Vpp
Fan Out	20 TTL load	20 TTL load
Rise or Fall Time	≤ 25ns	≤ 25ns
VCF (Voltage Controlled Frequency)		
Input Voltage	0V ~ 10V ±1V (100 : 1)	0V ~ 10V ±1V (100 : 1)
Input Impedance	10kΩ ±10%	10kΩ ±10%
GCV (Generator Controlled Voltage) (GFG-8219A/8255A)		
Output Voltage	sets the voltage between 0V ~ 2V	Sets the voltage between 0V ~ 2V
SWEEP OPERATION (GFG-8219A/8255A)		
Selection	Switch selector	Switch selector
Sweep/Rate	100 : 1 max, adjustable	100 : 1 max, adjustable
Sweep/Time	0.5sec. ~ 30sec. adjustable	0.5sec. ~ 30sec. adjustable
Sweep/Mode	Lin./Log. switch selector	Lin./Log. switch selector
AMPLITUDE MODULATION (GFG-8219A/8255A)		
Depth	0 ~ 100%	0 ~ 100%
MOD. Frequency	400Hz (INT), DC ~ 1MHz EXT	400Hz (INT), DC ~ 1MHz EXT
Carrier BW	100Hz ~ 3MHz (-3dB)	100Hz ~ 5MHz (-3dB)
EXT Sensitivity	≤ 10Vpp for 100% modulation	≤ 10Vpp for 100% modulation

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SPECIFICATIONS

MODEL	GFG-8216A/8219A	GFG-8255A
FREQUENCY MODULATION (GFG-8219A/8255A)		
Deviation	0 ~ ±5%	0 ~ ±5%
MOD. Frequency	400Hz(INT), DC ~ 20kHz(EXT)	400Hz (INT), DC ~ 20kHz(EXT)
EXT. Sensitivity	≤ 10Vpp for 10% modulation	≤ 10Vpp for 10% modulation
FREQUENCY COUNTER		
INT/EXT	Switch selector	Switch selector
Range	0.3Hz ~ 3MHz (5Hz ~ 150MHz EXT)	0.5Hz ~ 5MHz (5Hz ~ 150MHz EXT)
Accuracy	Time base accuracy ±1 count	Time base accuracy ±1 count
Time Base	± 20ppm (23 C±5 C) after 30 minutes warm up	± 20ppm (23 °C±5 °C) after 30 minutes warm up
Resolution	100nHz for 1Hz 1Hz for 100MHz	100nHz for 1Hz 1Hz for 100MHz
Input Impedance	1MΩ // 150pF	1MΩ // 150pF
Sensitivity	≤ 35mVrms (5Hz ~ 100MHz) ≤ 45mVrms (100MHz ~ 150MHz)	≤ 35mVrms (5Hz ~ 100MHz) ≤ 45mVrms (100MHz ~ 150MHz)
POWER SOURCE		
	AC115V, 230V±15%, 50 / 60Hz	AC115V, 230V±15%, 50 / 60Hz
DIMENSION & WEIGHT		
	251(W) x 91(H) x 291(D)mm Approx. 2.1 kg (GFG-8216A) Approx. 2.2 kg (GFG-8219A)	251(W) x 91(H) x 291(D)mm Approx. 2.4 kg (GFG-8255A)

ORDERING INFORMATION

GFG-8255A	5MHz Function Generator With Counter, Sweep Mode & AM/FM Modulation
GFG-8219A	3MHz Function Generator With Counter, Sweep Mode & AM/FM Modulation
GFG-8216A	3MHz Function Generator With Counter

ACCESSORIES :
User manual x 1; Test lead GTL-101 x 2

OPTIONAL ACCESSORIES

GTL-110	BNC Cable, BNC(P/M)-BNC(P/M), 1000mm

SELECTION GUIDE

MODEL	GFG-8255A	GFG-8219A	GFG-8216A
DISPLAY	6 Digits LED Display		
WAVEFORM	Sing , Square , Triangle		
FREQUENCY RANGE	5MHz	3MHz	3MHz
LIN/LOG SWEEP	✓	✓	
AM/FM MODULATION	✓	✓	
FREQUENCY COUNTER	✓	✓	✓
VOLTAGE CONTROL FREQUENCY	✓	✓	✓
GCV OUTPUT	✓	✓	
TTL/CMOS OUTPUT	✓	✓	✓

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